NAAN MUDHALVAN

LEVEL 2 – HACKATHON PROJECT REPORT

STUDENT ATTENDANCE TRACKER

Fundamentals of Web Development
HTML5 • CSS3 • JavaScript

PROJECT SUBMISSION

A browser-based application to manage student details, record daily attendance, calculate attendance percentages, search records and view attendance history.

GitHub Repository

https://github.com/saravananshiva975-collab/Saravanan-.git

Global College of Arts and Science
Submission date: 30.10.2026

STUDENT DETAILS

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Date of Submission	30.10.2026

GitHub Link (highlighted)

https://github.com/saravananshiva975-collab/Saravanan-.git

Project Overview

This report follows the structure of the supplied Naan Mudhalvan Level 2 model answer and documents the submitted Student Attendance Tracker source files.

1. PROBLEM STATEMENT

Keeping student attendance manually can be time-consuming and makes it difficult to find records, correct student details, and calculate attendance percentages. A simple browser-based tracker helps maintain student information and daily attendance in one place. The Student Attendance Tracker provides a single interface to add, edit and delete student records, mark attendance status, search and filter the table, and review each student's attendance history.

2. OBJECTIVES

• Add and maintain student ID, name and department.

• Mark daily attendance as Present, Absent, Late or Leave.

• Show summary counts for total students, present and absent students, and overall attendance.

• Search records by student name, ID or department.

• Filter the table by attendance status and set a minimum attendance threshold.

• View attendance history and remove a selected historical record.

• Store student and attendance data in browser Local Storage so it persists across reloads.

3. TECHNOLOGIES AND ARCHITECTURE

Technology	Role
HTML5	Defines the form, summary cards, controls, tables and history panel.
CSS3	Styles the dashboard, cards, buttons, tables and responsive layout.
JavaScript (ES6+)	Handles student management, attendance marking, validation, search, filters and history.
Web Storage API	Persists records in localStorage in the current browser.

Application flow

Open index.html → load saved data from Local Storage → add or select a student → mark daily status → save the record → refresh summary counts and attendance percentage → use search, status filter or History as needed.

4. FEATURES IMPLEMENTED

Student management: Add students with a unique ID; edit existing details and delete a student with their attendance records.

Daily attendance: Mark Present, Absent, Late or Leave for a selected date.

Summary dashboard: Displays total students, present today, absent today and overall attendance.

Search and filtering: Search by name, ID or department and filter by status.

Attendance threshold: Set a minimum percentage to identify low-attendance records.

History: View a student's attendance history and remove a selected historical entry.

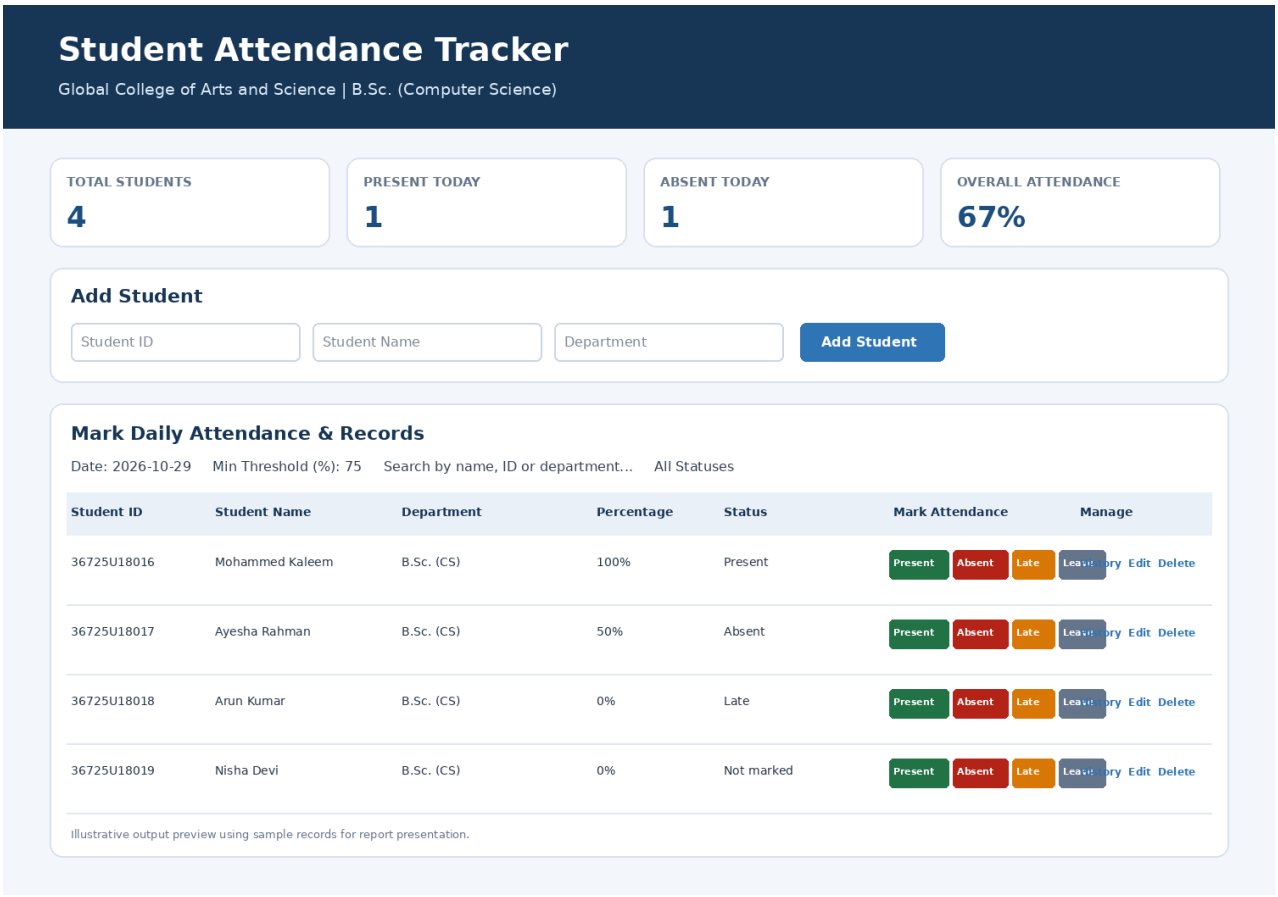
Local Storage: Saves the tracker data in the browser under the key attendanceTrackerData.

5. PROJECT EXECUTION

Extract the project ZIP file and keep index.html, style.css and script.js in the same folder. Open index.html in a modern browser such as Chrome, Edge, Firefox or Safari. Add student details, select a date, and use the status buttons to record attendance. The page updates the summary and student table after changes. The project does not require a server or package installation.

6. OUTPUT SCREENSHOT

The following report preview illustrates the application layout with sample student records and attendance statuses.



Note: the output image is a sample-data visual preview of the submitted interface, not a screenshot captured from a live browser session.

7. SOURCE CODE — HTML (index.html)

```
&lt;!DOCTYPE html&gt;
&lt;html lang="en"&gt;
&lt;head&gt;
  &lt;meta charset="UTF-8"&gt;
  &lt;meta name="viewport" content="width=device-width, initial-scale=1.0"&gt;
  &lt;title&gt;Student Attendance Tracker&lt;/title&gt;
  &lt;link rel="stylesheet" href="style.css"&gt;
&lt;/head&gt;
&lt;body&gt;
  &lt;header&gt;
    &lt;h1&gt;Student Attendance Tracker&lt;/h1&gt;
    &lt;p&gt;Global College of Arts and Science | B.Sc. (Computer Science)&lt;/p&gt;
  &lt;/header&gt;
  &lt;main&gt;
    &lt;section class="summary" aria-label="Attendance summary"&gt;
      &lt;article class="card"&gt;&lt;span&gt;Total Students&lt;/span&gt;&lt;strong id="total"&gt;0&lt;/
strong&gt;&lt;/article&gt;
      &lt;article class="card"&gt;&lt;span&gt;Present Today&lt;/span&gt;&lt;strong id="present"&gt;0&lt;/
strong&gt;&lt;/article&gt;
      &lt;article class="card"&gt;&lt;span&gt;Absent Today&lt;/span&gt;&lt;strong id="absent"&gt;0&lt;/
strong&gt;&lt;/article&gt;
      &lt;article class="card"&gt;&lt;span&gt;Overall Attendance&lt;/span&gt;&lt;strong id="overall"&gt;0%&lt;
/strong&gt;&lt;/article&gt;
    &lt;/section&gt;
    &lt;section class="panel"&gt;
      &lt;h2 id="formTitle"&gt;Add Student&lt;/h2&gt;
      &lt;form id="studentForm"&gt;
        &lt;input type="hidden" id="editOriginalId"&gt;
        &lt;input id="studentId" placeholder="Student ID (e.g., 36725U18016)" required&gt;
        &lt;input id="studentName" placeholder="Student Name" required&gt;
        &lt;input id="department" placeholder="Department (e.g., B.Sc. CS)" required&gt;
        &lt;button type="submit" id="submitBtn"&gt;Add Student&lt;/button&gt;
        &lt;button type="button" id="cancelEditBtn" class="btn-secondary hidden"&gt;Cancel&lt;/button&gt;
      &lt;/form&gt;
    &lt;/section&gt;
    &lt;section class="panel"&gt;
      &lt;h2&gt;Mark Daily Attendance &amp;amp; Records&lt;/h2&gt;
      &lt;div class="controls-bar"&gt;
        &lt;div class="control-group"&gt;&lt;label for="attendanceDate"&gt;Date:&lt;/label&gt;&lt;input
id="attendanceDate" type="date" required&gt;&lt;/div&gt;
        &lt;div class="control-group"&gt;&lt;label for="threshold"&gt;Min Threshold (%):&lt;/label&gt;&lt;
input id="threshold" type="number" min="0" max="100" value="75" class="small-input"&gt;&lt;/div&gt;
        &lt;div class="control-group grow"&gt;&lt;input id="search" placeholder="Search by name, ID or
department..."&gt;&lt;/div&gt;
        &lt;div class="control-group"&gt;&lt;select id="statusFilter"&gt;&lt;option value="All"&gt;All
Statuses&lt;/option&gt;&lt;option value="Present"&gt;Present&lt;/option&gt;&lt;option value="Absent"&gt;
Absent&lt;/option&gt;&lt;option value="Late"&gt;Late&lt;/option&gt;&lt;option value="Leave"&gt;Leave&lt;/
option&gt;&lt;option value="Not marked"&gt;Not Marked&lt;/option&gt;&lt;/select&gt;&lt;/div&gt;
      &lt;/div&gt;
      &lt;p id="message" role="status"&gt;&lt;/p&gt;
      &lt;div class="table-wrap"&gt;&lt;table&gt;&lt;thead&gt;&lt;tr&gt;&lt;th&gt;Student ID&lt;/th&gt;&lt;
th&gt;Student Name&lt;/th&gt;&lt;th&gt;Department&lt;/th&gt;&lt;th&gt;Percentage&lt;/th&gt;&lt;th&gt;Status (
Selected Date)&lt;/th&gt;&lt;th&gt;Mark Attendance&lt;/th&gt;&lt;th&gt;Manage&lt;/th&gt;&lt;/tr&gt;&lt;/
thead&gt;&lt;tbody id="studentRows"&gt;&lt;/tbody&gt;&lt;/table&gt;&lt;/div&gt;
    &lt;/section&gt;
    &lt;section id="historyPanel" class="panel hidden"&gt;
      &lt;div class="history-header"&gt;&lt;h2 id="historyTitle"&gt;Attendance History&lt;/h2&gt;&lt;button
type="button" id="closeHistoryBtn" class="btn-secondary"&gt;Close&lt;/button&gt;&lt;/div&gt;
      &lt;div class="table-wrap"&gt;&lt;table&gt;&lt;thead&gt;&lt;tr&gt;&lt;th&gt;Date&lt;/th&gt;&lt;th&gt;
Status&lt;/th&gt;&lt;th&gt;Action&lt;/th&gt;&lt;/tr&gt;&lt;/thead&gt;&lt;tbody id="historyRows"&gt;&lt;/
tbody&gt;&lt;/table&gt;&lt;/div&gt;
    &lt;/section&gt;
  &lt;/main&gt;
  &lt;script src="script.js"&gt;&lt;/script&gt;
&lt;/body&gt;
&lt;/html&gt;
```

8. SOURCE CODE — CSS (style.css)

```
* { box-sizing: border-box; }
body { margin: 0; font-family: Arial, sans-serif; background: #f3f6fb; color: #263548; }
header { background: #173656; color: white; padding: 24px 5%; }
header h1 { margin: 0 0 8px; }
header p { margin: 0; color: #dce9f8; }
main { max-width: 1200px; margin: 24px auto; padding: 18px; }
.summary { display: grid; grid-template-columns: repeat(4, 1fr); gap: 14px; margin-bottom: 20px; }
.card, .panel { background: white; border: 1px solid #d8e1ed; border-radius: 12px; padding: 18px; }
.card span { display: block; color: #65758b; font-size: 13px; text-transform: uppercase; }
.card strong { display: block; font-size: 28px; margin-top: 10px; color: #1c4e80; }
.panel { margin-bottom: 18px; }
.panel h2 { margin-top: 0; font-size: 18px; color: #173656; }
form, .controls-bar { display: flex; flex-wrap: wrap; gap: 12px; align-items: center; }
.control-group { display: flex; align-items: center; gap: 6px; }
.control-group.grow { flex: 1; }
.control-group.grow input { width: 100%; }
input, select, button { padding: 10px 12px; border: 1px solid #cbd6e4; border-radius: 7px; font-size: 14px; }
input { min-width: 150px; }
input.small-input { min-width: 70px; width: 75px; }
button { background: #2f75b5; color: white; border: 0; cursor: pointer; transition: background 0.2s; }
button:hover { background: #245d91; }
.btn-present { background: #217346; } .btn-present:hover { background: #195935; }
.btn-absent { background: #b42318; } .btn-absent:hover { background: #8c1b12; }
.btn-late { background: #d97706; } .btn-late:hover { background: #b45309; }
.btn-leave { background: #64748b; } .btn-leave:hover { background: #475569; }
.btn-secondary { background: #e2e8f0; color: #1e293b; } .btn-secondary:hover { background: #cbd5e1; }
.btn-danger { background: #fee2e2; color: #b42318; } .btn-danger:hover { background: #fecaca; }
.btn-sm { padding: 6px 10px; font-size: 12px; margin-right: 4px; }
.table-wrap { overflow-x: auto; margin-top: 14px; }
table { width: 100%; border-collapse: collapse; background: white; }
th, td { padding: 12px; text-align: left; border-bottom: 1px solid #e4eaf2; font-size: 14px; }
th { background: #eaf1f9; color: #173656; }
tr.low-row { background-color: #fffbfa; }
.status-present { color: #217346; font-weight: bold; }
.status-absent, .low { color: #b42318; font-weight: bold; }
.status-late { color: #d97706; font-weight: bold; }
.status-leave { color: #64748b; font-weight: bold; }
#message { min-height: 20px; margin: 10px 0 0; font-weight: bold; color: #217346; }
#message.error { color: #b42318; }
.hidden { display: none; }
.history-header { display: flex; justify-content: space-between; align-items: center; }
@media (max-width: 760px) { .summary { grid-template-columns: repeat(2, 1fr); } form, .controls-bar { flex-
direction: column; align-items: stretch; } input, select, button { width: 100%; } }
@media (max-width: 420px) { .summary { grid-template-columns: 1fr; } }
```

9. SOURCE CODE — JAVASCRIPT (script.js)

```
const KEY = "attendanceTrackerData";
const state = JSON.parse(localStorage.getItem(KEY) || '{"students": [], "attendance": []}');
const $ = id =&gt; document.getElementById(id);
const today = new Date().toISOString().slice(0, 10);
$("attendanceDate").value = today;
let activeHistoryStudentId = null;
function save() { localStorage.setItem(KEY, JSON.stringify(state)); }
function showMessage(text, isError = false) {
  const msgEl = $("message"); msgEl.textContent = text; msgEl.className = isError ? "error" : "";
}
function escapeHTML(value) {
  return String(value).replace(/[&amp;&lt;&gt;"']/g, ch =&gt; ({
    "&amp;": "&amp;amp;", "&lt;": "&amp;lt;", "&gt;": "&amp;gt;", '"': "&amp;quot;", "'": "&amp;#39;"
  }[ch]));
}
function saveStudent(originalId, studentId, name, department) {
  studentId = studentId.trim(); name = name.trim(); department = department.trim();
  if (!studentId || !name || !department) throw new Error("All student fields are required.");
  if (originalId) {
    if (originalId !== studentId &amp;&amp; state.students.some(s =&gt; s.studentId.toLowerCase() ===
studentId.toLowerCase())) throw new Error("New Student ID already exists.");
    const student = state.students.find(s =&gt; s.studentId === originalId);
    if (student) {
      student.studentId = studentId; student.name = name; student.department = department;
      state.attendance.forEach(a =&gt; { if (a.studentId === originalId) a.studentId = studentId; });
    }
  } else {
    if (state.students.some(s =&gt; s.studentId.toLowerCase() === studentId.toLowerCase())) throw new Error(
"Student ID already exists.");
    state.students.push({ studentId, name, department });
  }
  save();
}
function deleteStudent(studentId) {
  if (!confirm(`Delete student ${studentId} and all their attendance records?`)) return;
  state.students = state.students.filter(s =&gt; s.studentId !== studentId);
  state.attendance = state.attendance.filter(a =&gt; a.studentId !== studentId);
  if (activeHistoryStudentId === studentId) { $("historyPanel").classList.add("hidden");
activeHistoryStudentId = null; }
  save(); showMessage("Student deleted."); render();
}
function startEditStudent(studentId) {
  const student = state.students.find(s =&gt; s.studentId === studentId); if (!student) return;
  $("editOriginalId").value = student.studentId; $("studentId").value = student.studentId;
  $("studentName").value = student.name; $("department").value = student.department;
  $("formTitle").textContent = "Edit Student"; $("submitBtn").textContent = "Update Student";
  $("cancelEditBtn").classList.remove("hidden"); $("studentId").focus();
}
function resetForm() {
  $("studentForm").reset(); $("editOriginalId").value = "";
  $("formTitle").textContent = "Add Student"; $("submitBtn").textContent = "Add Student";
  $("cancelEditBtn").classList.add("hidden");
}
function markAttendance(studentId, date, status) {
  if (!date) throw new Error("Please select a valid date.");
  const existing = state.attendance.find(a =&gt; a.studentId === studentId &amp;&amp; a.date === date);
  if (existing) throw new Error(`Attendance already recorded as '${existing.status}' for ${date}. Delete it
from History to re-mark.`);
  state.attendance.push({ studentId, date, status }); save();
}
function percentage(studentId) {
  const rows = state.attendance.filter(a =&gt; a.studentId === studentId &amp;&amp; ["Present", "Absent"].
includes(a.status));
  if (!rows.length) return 0;
  return Math.round(rows.filter(a =&gt; a.status === "Present").length / rows.length * 100);
}
function getStatusClass(status) {
  if (status === "Present") return "status-present";
  if (status === "Absent") return "status-absent";
  if (status === "Late") return "status-late";
```

```
  if (status === "Leave") return "status-leave";
  return "";
}
function renderHistory(studentId) {
  const student = state.students.find(s =&gt; s.studentId === studentId); if (!student) return;
  activeHistoryStudentId = studentId;
  $("historyTitle").textContent = `Attendance History: ${student.name} (${student.studentId})`;
  const records = state.attendance.filter(a =&gt; a.studentId === studentId).sort((a,b) =&gt; b.date.
localeCompare(a.date));
  $("historyRows").innerHTML = !records.length ? `&lt;tr&gt;&lt;td colspan="3"&gt;No attendance records found.
&lt;/td&gt;&lt;/tr&gt;` :
    records.map(r =&gt; `&lt;tr&gt;&lt;td&gt;${escapeHTML(r.date)}&lt;/td&gt;&lt;td class="${getStatusClass(r.
status)}"&gt;${escapeHTML(r.status)}&lt;/td&gt;&lt;td&gt;&lt;button class="btn-danger btn-sm" data-del-
date="${escapeHTML(r.date)}" data-del-id="${escapeHTML(r.studentId)}"&gt;Remove Record&lt;/button&gt;&lt;/
td&gt;&lt;/tr&gt;`).join("");
  $("historyPanel").classList.remove("hidden");
}
function render() {
  const date = $("attendanceDate").value, q = $("search").value.toLowerCase().trim();
  const statusFilter = $("statusFilter").value, threshold = Number($("threshold").value) || 75;
  const filteredStudents = state.students.filter(s =&gt; {
    const matchesSearch = [s.studentId, s.name, s.department].join(" ").toLowerCase().includes(q);
    const saved = state.attendance.find(a =&gt; a.studentId === s.studentId &amp;&amp; a.date === date);
    const currentStatus = saved ? saved.status : "Not marked";
    return matchesSearch &amp;&amp; (statusFilter === "All" || currentStatus === statusFilter);
  });
  $("studentRows").innerHTML = !filteredStudents.length ? `&lt;tr&gt;&lt;td colspan="7"&gt;No matching
student records found.&lt;/td&gt;&lt;/tr&gt;` :
    filteredStudents.map(s =&gt; {
      const saved = state.attendance.find(a =&gt; a.studentId === s.studentId &amp;&amp; a.date === date);
      const status = saved ? saved.status : "Not marked";
      const hasRecords = state.attendance.some(a =&gt; a.studentId === s.studentId &amp;&amp; ["Present",
"Absent"].includes(a.status));
      const isLow = hasRecords &amp;&amp; pct &lt; threshold;
      return `&lt;tr class="${isLow ? 'low-row' : ''}"&gt;
        &lt;td&gt;${escapeHTML(s.studentId)}&lt;/td&gt;&lt;td&gt;${escapeHTML(s.name)}&lt;/td&gt;&lt;td&gt;${
escapeHTML(s.department)}&lt;/td&gt;
        &lt;td class="${isLow ? 'low' : 'status-present'}"&gt;${pct}% ${isLow ? '(Low)' : ''}&lt;/td&gt;
        &lt;td class="${getStatusClass(status)}"&gt;${escapeHTML(status)}&lt;/td&gt;
        &lt;td&gt;&lt;button class="btn-present btn-sm" data-action="mark" data-id="${escapeHTML(s.
studentId)}" data-status="Present"&gt;Present&lt;/button&gt;
        &lt;button class="btn-absent btn-sm" data-action="mark" data-id="${escapeHTML(s.studentId)}" data-
status="Absent"&gt;Absent&lt;/button&gt;
        &lt;button class="btn-late btn-sm" data-action="mark" data-id="${escapeHTML(s.studentId)}" data-
status="Late"&gt;Late&lt;/button&gt;
        &lt;button class="btn-leave btn-sm" data-action="mark" data-id="${escapeHTML(s.studentId)}" data-
status="Leave"&gt;Leave&lt;/button&gt;&lt;/td&gt;
        &lt;td&gt;&lt;button class="btn-secondary btn-sm" data-action="history" data-id="${escapeHTML(s.
studentId)}"&gt;History&lt;/button&gt;
        &lt;button class="btn-secondary btn-sm" data-action="edit" data-id="${escapeHTML(s.studentId)}"&gt;
Edit&lt;/button&gt;
        &lt;button class="btn-danger btn-sm" data-action="delete" data-id="${escapeHTML(s.studentId)}"&gt;
Delete&lt;/button&gt;&lt;/td&gt;&lt;/tr&gt;`;
    }).join("");
  const todays = state.attendance.filter(a =&gt; a.date === date);
  $("total").textContent = state.students.length;
  $("present").textContent = todays.filter(a =&gt; a.status === "Present").length;
  $("absent").textContent = todays.filter(a =&gt; a.status === "Absent").length;
  const percentages = state.students.map(s =&gt; percentage(s.studentId));
  $("overall").textContent = percentages.length ? Math.round(percentages.reduce((a,b) =&gt; a+b,0)/
percentages.length)+"%" : "0%";
  if (activeHistoryStudentId) renderHistory(activeHistoryStudentId);
}
$("studentForm").addEventListener("submit", event =&gt; {
  event.preventDefault();
  try {
    const origId = $("editOriginalId").value;
    saveStudent(origId, $("studentId").value, $("studentName").value, $("department").value);
    showMessage(origId ? "Student updated successfully." : "Student added successfully."); resetForm();
render();
  } catch (error) { showMessage(error.message, true); }
});
$("cancelEditBtn").addEventListener("click", resetForm);
$("studentRows").addEventListener("click", event =&gt; {
  const button = event.target.closest("button[data-action]"); if (!button) return;
  const { action, id, status } = button.dataset;
  try {
    if (action === "mark") { markAttendance(id, $("attendanceDate").value, status); showMessage(`Marked ${
status} for ${id}.`); render(); }
    else if (action === "edit") startEditStudent(id);
    else if (action === "delete") deleteStudent(id);
    else if (action === "history") renderHistory(id);
  } catch (error) { showMessage(error.message, true); }
```

```
});
$("historyRows").addEventListener("click", event =&gt; {
  const button = event.target.closest("button[data-del-date]"); if (!button) return;
  const { delId, delDate } = button.dataset;
  state.attendance = state.attendance.filter(a =&gt; !(a.studentId === delId &amp;&amp; a.date === delDate));
  save(); showMessage(`Removed attendance record for ${delDate}.`); render();
});
$("closeHistoryBtn").addEventListener("click", () =&gt; { $("historyPanel").classList.add("hidden");
activeHistoryStudentId = null; });
$("search").addEventListener("input", render); $("statusFilter").addEventListener("change", render);
$("threshold").addEventListener("input", render); $("attendanceDate").addEventListener("change", render);
render();
```

10. CONCLUSION

The Student Attendance Tracker demonstrates how HTML, CSS and JavaScript can be combined to build a useful browser-based attendance tool. It supports student record management, daily attendance marking, summary calculations, search and filtering, and attendance history. Local Storage provides persistence within the browser. Future improvements could include exporting attendance reports, importing student lists, login roles and a shared database for multiple devices.

Repository

https://github.com/saravananshiva975-collab/Saravanan-.git